

Climate 5000 Series - Gen 4

12K BTU Single Zone Max Performance System

1-Way Cassette IDU

Engineering
Submittal
Sheet



BOSCH

Overview and Project Information



Project name

Location

Architect

Engineer

General contractor

Reference #

12K Btu Single Zone Max Performance System - 1-Way Cassette IDU

Indoor part number	8-733-962-667
Indoor model number	BMS500-AAU012-1AHZXD
Outdoor part number	8-733-962-689
Outdoor model number	BMS500-AAS012-1CSXHD

Efficiency Data		AHRI 210/240 2023
SEER2		23
EER2		13
HSPF2 Region IV		10
HSPF2 Region V		8
COP(47F)		3.56
COP (5F)		2.0

Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	4000-13900
	Rated Capacity	12000
	Current	4.3
	Input Power	923
Heating	Capacity Range	4600-15200
	Rated Capacity (47F)	12500
	Rated Capacity (5F)	11500
	Current	4.6
	Input Power	1030

Electrical Data	
Power supply (V / Ø / Hz)	208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)	3.0 / 15.0
Max. fuse indoor/outdoor (A)	15.0 / 15.0

Compressor Data	
Type:	Rotary
Refrigerant oil /oil charge (ml)	VG74 / 350
Input (W)	885
Rated current (RLA/A)	6.0

Piping	
Refrigerant precharge (oz/kg)	35.27 (1.00)
Length for precharge (ft/m)	24.6 / 7.5
Liquid side (in/mm)	1/4" / Ø6.35
Gas side (in/mm)	3/8" / Ø9.52
Additional Charge of Refrigerant (oz/ft)/(g/m)	0.16 / 15
Max. refrigerant pipe length (ft/m)	82.02 / 25
Max. difference in height (ft/m)	49.21 / 15

Operation Range

ODU	Cooling (°F / °C)	-22~122 / -30~50
	Heating (°F / °C)	-22~75 / -30~24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

System Data

Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		3.84
IDU	Fan motor model	ZKFN-30-8-43
	Fan motor qty (ea)	1
	Fan motor input (W)	46.0
	Fan speed (RPM)	1184 / 1064 / 944
	Air flow, Turbo / H / M / L / Si (CFM)	353.2 / 312.0 / 282.5 / 247.2 / 176.6
	Approx. Air Throw Range ft (m)	23.0 (7)
	Noise level, Turbo / H / M / L / Si (dB(A))	N/A / 41 / 37 / 33 / 24
	Net weight (lbs / kg)	45.19 / 20.5
ODU	Gross weight (lbs / kg)	82.45 / 37.4
	Drain pipe diameter (mm)	ODØ25mm
	Fan motor model	ZKFN-34-10-1-3
	Fan motor qty (ea)	1
	Fan motor input (W)	50
	Fan motor speed (RPM)	780 / 680 / 580
	Air flow, Max (CFM)	1471.5
	Noise level, Max (dB(A))	56
	Net weight (lbs / kg)	72.75 / 33
	Gross weight (lbs / kg)	78.7 / 35.7

Accessories

BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
BMS-DRY-XXD (8733969263)	24V Interface
BMS-PK2-7X1 (8733951011)	25ft Lineset

Standard Parts

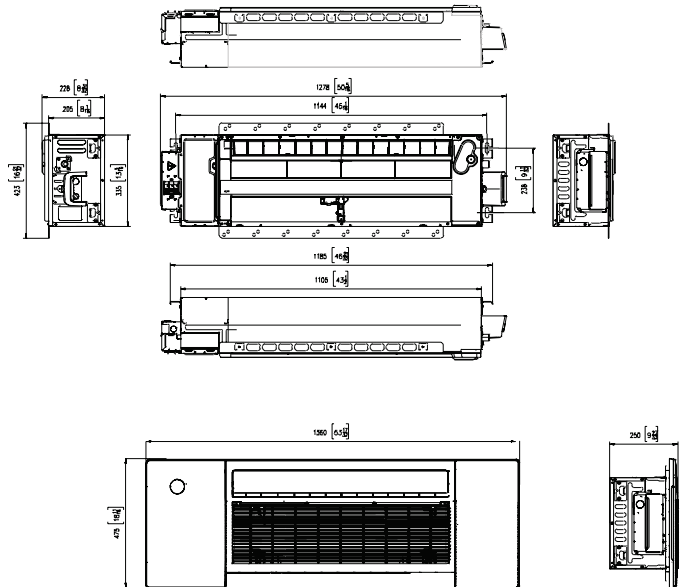
8-733-969-065	Remote Control
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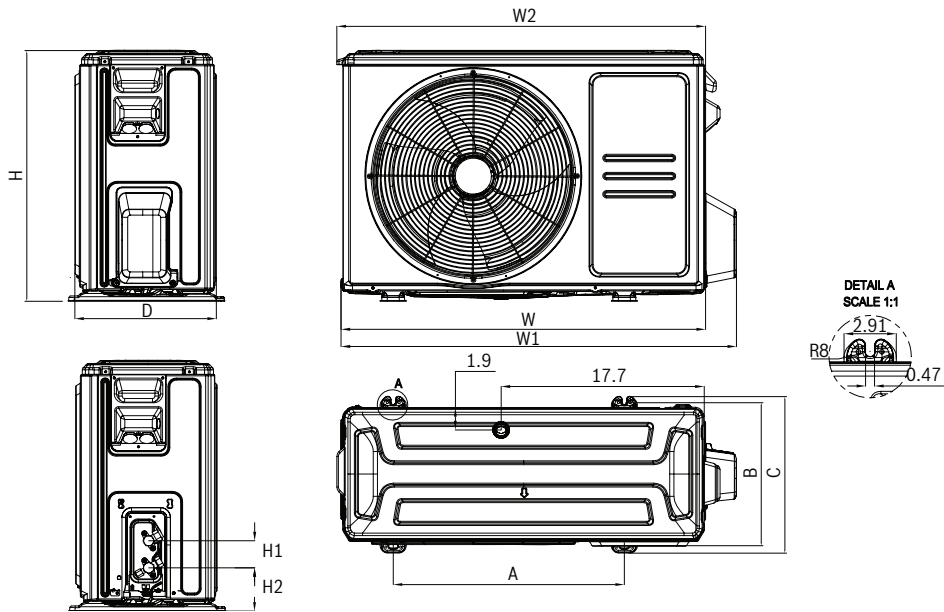


Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU012-1AHZXD	50.31	13.19	8.98

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	A (in)	B (in)	C (in)	H1 (in)	H2 (in)
BMS500-AAS012-1CSXHD	32	13	21.8	34.4	32.1	20.1	12.5	13.6	2.4	3.8

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Cooling Performance - Indoor Dry Bulb Temperature at 353 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F (16.0°C)					64.4°F (18.0°C)					67°F (19.4°C)					71.6°F (22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-22°F (-30.0°C)	TC	9849	9845	9985	10026	10302	10276	10268	10261	10260	10252	10583	10570	10560	10559	10548	11163	11160	11137	11133	11110
	S/T	0.80	0.91	0.99	0.99	0.99	0.64	0.75	0.84	0.86	0.97	0.55	0.65	0.74	0.75	0.86	0.39	0.47	0.55	0.57	0.67
	PI	0.51	0.51	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.51	0.51	0.51	0.51	0.51
-13°F (-25.0°C)	TC	10670	10665	10817	10862	11161	11132	11123	11116	11115	11106	11465	11451	11440	11438	11427	12094	12090	12066	12061	12035
	S/T	0.80	0.91	1.00	1.00	1.00	0.65	0.75	0.85	0.86	0.97	0.55	0.65	0.74	0.76	0.86	0.39	0.47	0.56	0.57	0.67
	PI	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52
-4°F (-20.0°C)	TC	11491	11485	11649	11697	12019	11989	11979	11971	11970	11960	12347	12332	12320	12318	12306	13024	13020	12994	12989	12961
	S/T	0.80	0.92	1.00	1.00	1.00	0.65	0.76	0.85	0.87	0.98	0.55	0.65	0.74	0.76	0.86	0.39	0.48	0.56	0.57	0.67
	PI	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54
0°F (-17.8°C)	TC	11872	11865	12085	12136	12469	12381	12367	12358	12356	12481	12745	12727	12711	12709	12703	13430	13427	13426	13428	13437
	S/T	0.81	0.93	1.00	1.00	1.00	0.66	0.77	0.86	0.88	1.00	0.55	0.66	0.75	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55	0.55
5°F (-15.0°C)	TC	12212	12254	12534	12592	12980	12785	12788	12789	12802	12993	13194	13197	13200	13200	13218	13968	13964	13971	13972	13978
	S/T	0.82	0.94	1.00	1.00	1.00	0.66	0.78	0.87	0.88	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.59	0.69
	PI	0.55	0.55	0.55	0.55	0.56	0.55	0.55	0.55	0.55	0.56	0.56	0.56	0.56	0.56	0.56	0.57	0.57	0.57	0.57	0.57
14°F (-10.0°C)	TC	11580	11574	11811	11866	12234	12155	12144	12136	12135	12248	12572	12556	12543	12541	12528	13368	13363	13338	13333	13305
	S/T	0.81	0.93	1.00	1.00	1.00	0.65	0.76	0.86	0.87	1.00	0.55	0.66	0.75	0.76	0.87	0.39	0.48	0.56	0.57	0.67
	PI	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56
17°F (-8.3°C)	TC	11344	11381	11644	11700	12080	11908	11913	11915	11925	12093	12315	12319	12324	12325	12340	13097	13093	13091	13094	13105
	S/T	0.82	0.93	1.00	1.00	1.00	0.66	0.77	0.87	0.88	1.00	0.55	0.66	0.76	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56
23°F (-5.0°C)	TC	11419	11468	11765	11823	12213	11997	12001	12002	12018	12226	12411	12415	12419	12419	12439	13197	13193	13202	13204	13211
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.89	0.39	0.48	0.57	0.59	0.69
	PI	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.57	0.57	0.57	0.57	0.57	0.58	0.58	0.58	0.58	0.58
35°F (1.7°C)	TC	11679	11679	11748	11781	12161	12225	12224	12225	12225	12258	12608	12607	12607	12608	12609	13317	13314	13314	13313	13314
	S/T	0.78	0.88	0.97	0.99	1.00	0.63	0.73	0.82	0.83	0.93	0.54	0.63	0.72	0.73	0.83	0.39	0.46	0.54	0.56	0.65
	PI	0.52	0.52	0.53	0.53	0.55	0.56	0.56	0.56	0.56	0.56	0.58	0.58	0.58	0.58	0.58	0.63	0.63	0.63	0.63	0.63
41°F (5.0°C)	TC	11632	11632	11725	11806	12155	12173	12173	12173	12173	12216	12553	12552	12553	12553	12554	13257	13254	13253	13252	13254
	S/T	0.78	0.89	0.98	1.00	1.00	0.64	0.74	0.83	0.84	0.94	0.54	0.64	0.72	0.74	0.84	0.39	0.47	0.55	0.56	0.66
	PI	0.55	0.55	0.55	0.56	0.58	0.58	0.58	0.58	0.58	0.59	0.61	0.61	0.61	0.61	0.61	0.66	0.66	0.66	0.66	0.66
47°F (8.3°C)	TC	11585	11592	11748	11800	12150	12122	12122	12122	12122	12178	12498	12497	12498	12498	12498	13198	13195	13194	13193	13194
	S/T	0.79	0.90	1.00	1.00	1.00	0.64	0.75	0.84	0.85	0.96	0.54	0.65	0.73	0.75	0.85	0.39	0.47	0.55	0.57	0.67
	PI	0.57	0.57	0.58	0.59	0.61	0.61	0.61	0.61	0.61	0.62	0.64	0.64	0.64	0.64	0.64	0.70	0.70	0.70	0.70	0.70
50°F (10.0°C)	TC	11389	11702	12097	12167	12639	12038	12038	12121	12160	12651	12501	12517	12524	12524	12645	13363	13398	13432	13436	13453
	S/T	0.90	1.00	1.00	1.00	1.00	0.71	0.85	0.96	0.99	1.00	0.59	0.72	0.83	0.85	0.99	0.40	0.50	0.60	0.62	0.74
	PI	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64
59°F (15.0°C)	TC	11422	11823	12242	12317	12818	12085	12101	12239	12330	12831	12583	12596	12600	12619	12841	13512	13539	13566	13569	13580
	S/T	0.92	1.00	1.00	1.00	1.00	0.73	0.86	1.00	1.00	1.00	0.60	0.73	0.85	0.86	1.00	0.40	0.51	0.61	0.63	0.76
	PI	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.66	0.66	0.66	0.66	0.66
68°F (20.0°C)	TC	11449	11938	12381	12460	12988	12121	12148	12394	12473	13001	12645	12654	12677	12688	13012	13625	13650	13672	13674	13681
	S/T	0.95	1.00	1.00	1.00	1.00	0.74	0.89	1.00	1.00	1.00	0.61	0.75	0.87	0.89	1.00	0.40	0.51	0.63	0.65	0.78
	PI	0.65	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.67	0.66	0.66	0.66	0.66	0.67	0.67	0.67	0.67	0.67	0.67
77°F (25.0°C)	TC	11467	12019	12490	12573	13135	12157	12195	12503	12587	13148	12707	12721	12754	12768	13159	13724	13766	13798	13801	13814
	S/T	0.97	1.00	1.00	1.00	1.00	0.75	0.90	1.00	1.00	1.00	0.61	0.76	0.88	0.90	1.00	0.40	0.52	0.63	0.65	0.79
	PI	0.66	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.68	0.68	0.68	0.68	0.68
82°F (27.8°C)	TC	11604	12197	12689	12775	13360	12297	12345	12700	12789	13374	12869	12885	12925	12941	13386	13914	13967	14003	14007	14022
	S/T	0.99	1.00	1.00	1.00	1.00	0.76	0.91	1.00	1.00	1.00	0.62	0.77	0.89	0.92	1.00	0.40	0.52	0.64	0.66	0.80
	PI	0.74	0.74	0.75	0.75	0.75	0.74	0.74	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.76	0.76	0.76	0.76	0.75
86°F (30.0°C)	TC	11916	12511	13023	13113	13724	12566	12624	13037	13128	13738	13162	13177	13225	13244	13750	14249	14304	14338	14342	14355
	S/T	1.00	1.00	1.00	1.00	1.00	0.77	0.93	1.00	1.00	1.00	0.62	0.78	0.91	0.93	1.00	0.40	0.52	0.64	0.66	0.81
	PI	0.88	0.89	0.89	0.90	0.90	0.89	0.89	0.89	0.89	0.90	0.90	0.90	0.90	0.90	0.90	0.91	0.91	0.91	0.91	0.91
95°F (35.0°C)	TC	12530	12598	13005	13082	13599	13317	13325	13												

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Heating Performance - Indoor Dry Bulb Temperature at 353 CFM										
Outdoor DB(°F)	TC Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	5230	5126	5100	5080	5034	1.36	1.37	1.37	1.37	1.38
-13°F(-25.0°C)	6461	6332	6300	6275	6218	1.43	1.44	1.44	1.45	1.45
-4°F(-20.0°C)	8820	8643	8600	8566	8489	1.52	1.53	1.54	1.54	1.55
0°F(-17.8°C)	9731	9537	9489	9451	9366	1.59	1.61	1.61	1.61	1.62
5°F(-15.0°C)	10871	10653	10600	10558	10463	1.68	1.69	1.70	1.70	1.71
14°F(-10.0°C)	11871	11633	11575	11529	11425	1.75	1.76	1.77	1.77	1.78
17°F(-8.3°C)	12204	11960	11900	11852	11746	1.77	1.78	1.79	1.79	1.80
22°F(-5.6°C)	12572	12320	12259	12210	12100	1.76	1.78	1.78	1.78	1.79
27°F(-2.8°C)	12939	12681	12617	12567	12454	1.75	1.77	1.77	1.78	1.78
32°F(0.0°C)	13307	13041	12976	12924	12807	1.75	1.76	1.76	1.77	1.77
35°F(1.7°C)	13528	13257	13191	13138	13020	1.74	1.76	1.76	1.76	1.77
37°F(2.8°C)	13871	13594	13526	13472	13350	1.68	1.69	1.69	1.70	1.71
42°F(5.6°C)	14730	14435	14363	14305	14177	1.52	1.53	1.54	1.54	1.55
44.6°F(7.0°C)	15073	14772	14698	14639	14507	1.46	1.47	1.47	1.48	1.48
47°F(8.3°C)	15588	15276	15200	15139	15003	1.36	1.37	1.38	1.38	1.39
52°F(11.1°C)	15651	15338	15261	15200	15063	1.20	1.21	1.21	1.22	1.22
57°F(13.9°C)	15713	15399	15322	15261	15123	1.04	1.05	1.05	1.05	1.05
59°F(15.0°C)	15738	15423	15346	15285	15147	0.97	0.98	0.98	0.98	0.99
62°F(16.7°C)	15776	15460	15383	15321	15183	0.88	0.88	0.88	0.89	0.89
64.4°F(18.0°C)	15806	15489	15412	15350	15212	0.80	0.80	0.81	0.81	0.81
69.8°F(21.0°C)	15873	15556	15478	15416	15277	0.62	0.63	0.63	0.63	0.63
75.2°F(24.0°C)	15941	15622	15544	15481	15342	0.45	0.45	0.45	0.45	0.45

Bolded values were tested. All other values are simulated and for reference only.